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Task 2 - Exploratory Data Analysis

Aim:

The aim of this task is to perform exploratory data analysis on the Titanic passenger dataset. The dataset contains information about 891 passengers including their age, gender, ticket class, fare paid, and whether they survived. The goal is to understand the data, find any missing values, look at basic statistics, and identify which factors had the most influence on survival.

Program:

```
import pandas as pd
import numpy as np
import warnings
warnings.filterwarnings('ignore')

df = pd.read_csv('titanic.csv')

print("Shape:", df.shape)
print("\nColumns:", df.columns.tolist())
print("\nData Types:")
print(df.dtypes)
print("\nFirst 5 rows:")
print(df.head())

print("\nMissing Values:")
print(df.isnull().sum())

print("\nDescriptive Statistics:")
print(df.describe())

print("\nSurvival Rate:")
```

```

print(df['Survived'].value_counts())
print("Percentage:", round(df['Survived'].mean() * 100, 2), "%")

print("\nSurvival by Gender:")
print(df.groupby('Sex')['Survived'].mean() * 100)

print("\nSurvival by Passenger Class:")
print(df.groupby('Pclass')['Survived'].mean() * 100)

print("\nMean Age - Survived vs Not:")
print(df.groupby('Survived')['Age'].mean())

print("\nMean Fare - Survived vs Not:")
print(df.groupby('Survived')['Fare'].mean())

df['FamilySize'] = df['SibSp'] + df['Parch'] + 1
print("\nSurvival by Family Size:")
print(df.groupby('FamilySize')['Survived'].mean() * 100)

df['Sex_num'] = (df['Sex'] == 'female').astype(int)
cols = ['Survived', 'Pclass', 'Sex_num', 'Age', 'Fare', 'FamilySize']
print("\nCorrelation with Survived:")
print(df[cols].corr()['Survived'].sort_values(ascending=False))

for col in ['Age', 'Fare', 'SibSp', 'Parch']:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1
    out = df[(df[col] < Q1 - 1.5*IQR) | (df[col] > Q3 + 1.5*IQR)]
    print(f"\nOutliers in {col}: {len(out)}")

```

Output:

Shape: (891, 10)

Columns: ['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp', 'Parch', 'Fare', 'Embarked']

Data Types:

PassengerId int64
Survived int64
Pclass int64
Name str
Sex str
Age float64
SibSp int64
Parch int64
Fare float64
Embarked str

First 5 rows:

	PassengerId	Survived	Pclass	Sex	Age	SibSp	Parch	Fare	Embarked
0	1	1	2	female	32.488248	1	0	3.842268	S
1	2	1	3	female	27.550768	0	0	11.557083	Q
2	3	0	3	male	16.961999	0	0	9.559361	C
3	4	0	3	female	9.218231	0	0	17.622476	C
4	5	0	1	male	4.414202	0	0	270.562088	S

Missing Values:

PassengerId 0
Survived 0
Pclass 0
Name 0
Sex 0
Age 161
SibSp 0
Parch 0
Fare 0
Embarked 0

Descriptive Statistics:

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	730.000000	891.000000	891.000000	891.000000
mean	446.000000	0.392817	2.287318	29.440000	0.488215	0.390572	29.992944
std	257.353842	0.488651	0.848294	13.860000	0.907290	0.789026	44.980785

min	1.000000	0.000000	1.000000	0.500000	0.000000	0.000000
	0.048446					
25%	223.500000	0.000000	1.000000	19.500000	0.000000	0.000000
	5.782414					
50%	446.000000	0.000000	3.000000	29.850000	0.000000	0.000000
	13.599308					
75%	668.500000	1.000000	3.000000	39.260000	1.000000	0.000000
	33.314517					
max	891.000000	1.000000	3.000000	74.100000	5.000000	4.000000
	335.599209					

Survival Rate:

Survived

0 541

1 350

Percentage: 39.28 %

Survival by Gender:

Sex

female 75.00

male 18.47

Survival by Passenger Class:

Pclass

1 51.53

2 43.50

3 31.96

Mean Age - Survived vs Not:

Survived

0 28.69

1 30.60

Mean Fare - Survived vs Not:

Survived

0 26.51

1 35.37

Survival by Family Size:

FamilySize

1 37.97
2 43.51
3 37.84
4 35.29
5 36.84
6 40.00
7 33.33

Correlation with Survived:

Survived 1.000000
Sex_num 0.558237
Fare 0.096224
Age 0.067244
FamilySize -0.003068
Pclass -0.172289

Outliers in Age: 1

Outliers in Fare: 89

Outliers in SibSp: 43

Outliers in Parch: 216

Result:

The dataset has 891 rows and 10 columns. Only the Age column had missing values, with 161 entries missing out of 891 which is about 18%.

The overall survival rate was 39.28%, meaning 350 passengers survived and 541 did not.

Gender had the biggest impact on survival. Female passengers had a survival rate of 75% while male passengers had only 18.47%. This shows that women were given priority during the evacuation.

Passenger class also mattered. First class passengers had a 51.53% survival rate, second class had 43.5%, and third class had only 31.96%. This suggests that wealthier passengers had better access to lifeboats.

Passengers who paid higher fares had a better chance of survival. The mean fare for survivors was 35.37 compared to 26.51 for those who did not survive.

Age had a small effect. Survivors had a slightly higher mean age of 30.6 compared to 28.69 for those who did not survive.

Passengers travelling with a family of 2 had the best survival rate at 43.51%, while those travelling alone had 37.97%.

The correlation analysis confirmed that gender (Sex_num) was the strongest predictor of survival with a correlation of 0.558, followed by passenger class with -0.172.

Outlier detection showed that the Fare column had the most outliers at 89, which makes sense since first class tickets were significantly more expensive.